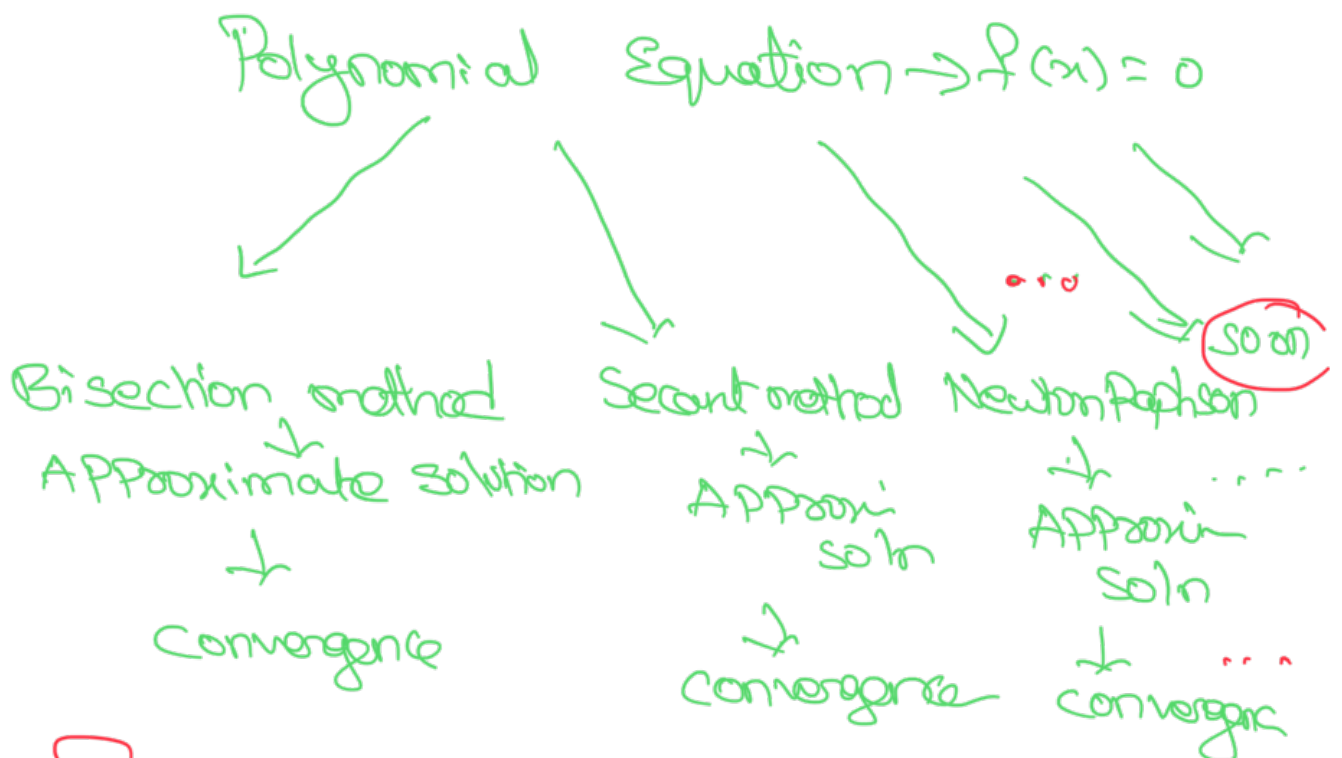
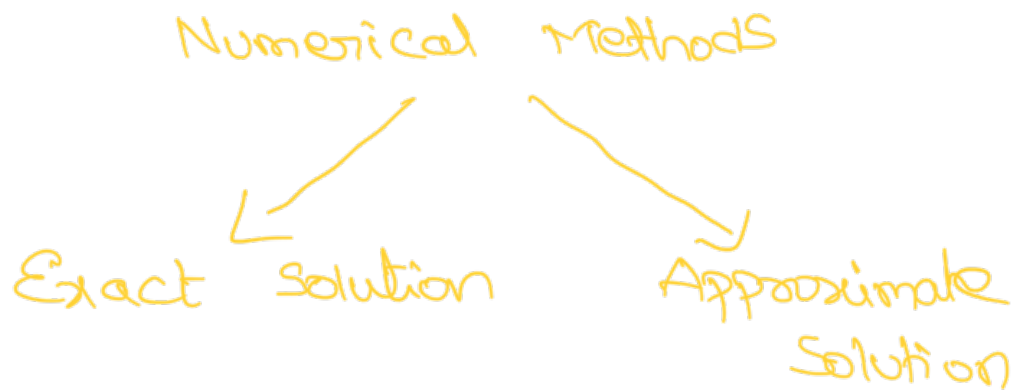
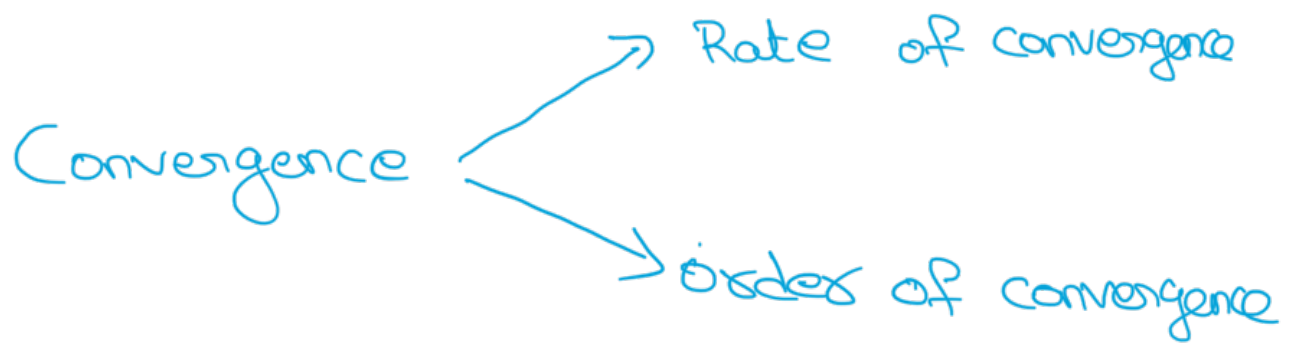




Faster convergence $\Rightarrow$ Good Method

Convergence of a sequence

$$\{x_n\} \longrightarrow L$$

$$\lim_{n \rightarrow \infty} x_n = L$$

(or)

$$\lim_{n \rightarrow \infty} |x_n - L| = 0$$

Eg: $\{\frac{1}{n}\} = \{1, \frac{1}{2}, \frac{1}{3}, \dots\}$

$$\{\frac{1}{n}\} \xrightarrow{\text{as } n \rightarrow \infty} 0$$

Rate of Convergence:-

Let $\{P_n\}$ be a sequence that converges to a number P . If there exists a sequence $\{P_n\}$ that converges to zero and a Positive constant λ , independent of n , such that

$$|P_n - P| \leq \lambda |P_n|$$

for all sufficiently large values of n , then $\{P_n\}$ is said to converge to P with RATE OF CONVERGENCE $O(P_n)$

$$P_n = P + O(B_n)$$

i.e
In short $\{P_n\} \rightarrow P \quad \exists \quad \{B_n\} \rightarrow 0 \text{ and } \lambda > 0$
s.t. $|P_n - P| \leq \lambda |B_n|$
 $\downarrow$
+ve constant

Eg 1:- $\left\{ \frac{n+3}{n+7} \right\} \rightarrow 1$ with R.O.C $O\left(\frac{1}{n}\right)$
 $\hookrightarrow P_n$ $\downarrow$ P Rate of convergence



Consider $\left| \frac{n+3}{n+7} - 1 \right| = \left| \frac{n+3-n-7}{n+7} \right|$

A hand-drawn diagram consisting of a large circle. Inside the circle, there are several lines and symbols. At the top, there is a horizontal line with a small 'u' above it. Below this, there are two vertical lines, each with a small 'u' to its right. In the center, there is a horizontal line with a small 'u' above it. At the bottom, there is a horizontal line with a small 'u' above it. The entire diagram is drawn in a single continuous line.

$$= \frac{-4}{177}$$

$$= \frac{4}{7+7} < \frac{4}{7} < \frac{4}{6}$$

$$\text{at } n \rightarrow \infty \quad P_n \rightarrow 0$$

$\therefore R.O.C \text{ is } \underline{O\left(\frac{1}{n}\right)}$

Eg 2:- $\left\{ \frac{2^n + 3}{2^n + 7} \right\} \rightarrow 1$ with R.O.C $\mathcal{O}\left(\frac{1}{2^n}\right)$



Consider,

② $\{P_n\} \begin{cases} \frac{1}{n^a} \\ \frac{1}{a^n} \end{cases}$ a is the const

③ $P_n = P + O(P_n)$

Error (e_n)

$e_n = P_n - P$

✓✓

Order of Convergence

↳ measure of effectiveness with which each iteration reduces the approximation error.

Definition! - Let $\{P_n\}$ be a sequence that converges to a number P .

Let $e_n = P_n - P$ for $n \geq 0$. If

$\exists$ Positive constant λ and α

such that $\lim_{n \rightarrow \infty} \frac{|P_{n+1} - P|}{|P_n - P|^\alpha} = \lambda$

(or)

$$\lim_{n \rightarrow \infty} \frac{|e_{n+1}|}{|e_n|^\alpha} = \lambda$$

then $\{P_n\}$ is said to
converge to P of ORDER ' α '
with ASYMPTOTIC ERROR
CONSTANT ' λ '

$\alpha = 1 \rightarrow$ linear

$\alpha = 2 \rightarrow$ Quadratic

$\alpha = 3 \rightarrow$ Cubic.

In Order of convergence

Quadratic $\Rightarrow$ Good method.

Even cubic is good but

for computing it takes extra work to perform. So quadratic is good method.

eg: $x_{n+1} = \frac{1}{2} \left(x_n + \frac{a}{x_n} \right)$
 $\{x_n\} \rightarrow \sqrt{a}, a \in \mathbb{R}^+$

↓ If we have sequence & this we get $\{x_n\} \rightarrow \sqrt{a}$

$$\lim_{n \rightarrow \infty} \frac{|e_{n+1}|}{|e_n|^d} = \lim_{n \rightarrow \infty} \frac{|p_{n+1} - p|}{|p_n - p|^d}$$

$$= \lim_{n \rightarrow \infty} \frac{|x_{n+1} - \sqrt{a}|}{|x_n - \sqrt{a}|^d}$$

$$= \lim_{n \rightarrow \infty} \frac{(x_n - \sqrt{a})^2}{|x_n - \sqrt{a}|^d} \cdot \frac{1}{2x_n}$$

$$= \lim_{n \rightarrow \infty} \underbrace{\frac{1}{2x_n}}_{\text{for } d=2} = \frac{1}{2\sqrt{a}} \downarrow \text{const}$$

for $d=2$

$$\begin{aligned} x_{n+1} - \sqrt{a} &= \frac{1}{2} \left(x_n + \frac{a}{x_n} \right) - \sqrt{a} \\ &= \frac{x_n^2 + a - 2x_n\sqrt{a}}{2x_n} \\ x_{n+1} - \sqrt{a} &= \frac{(x_n - \sqrt{a})^2}{2x_n} \end{aligned}$$

here, ~

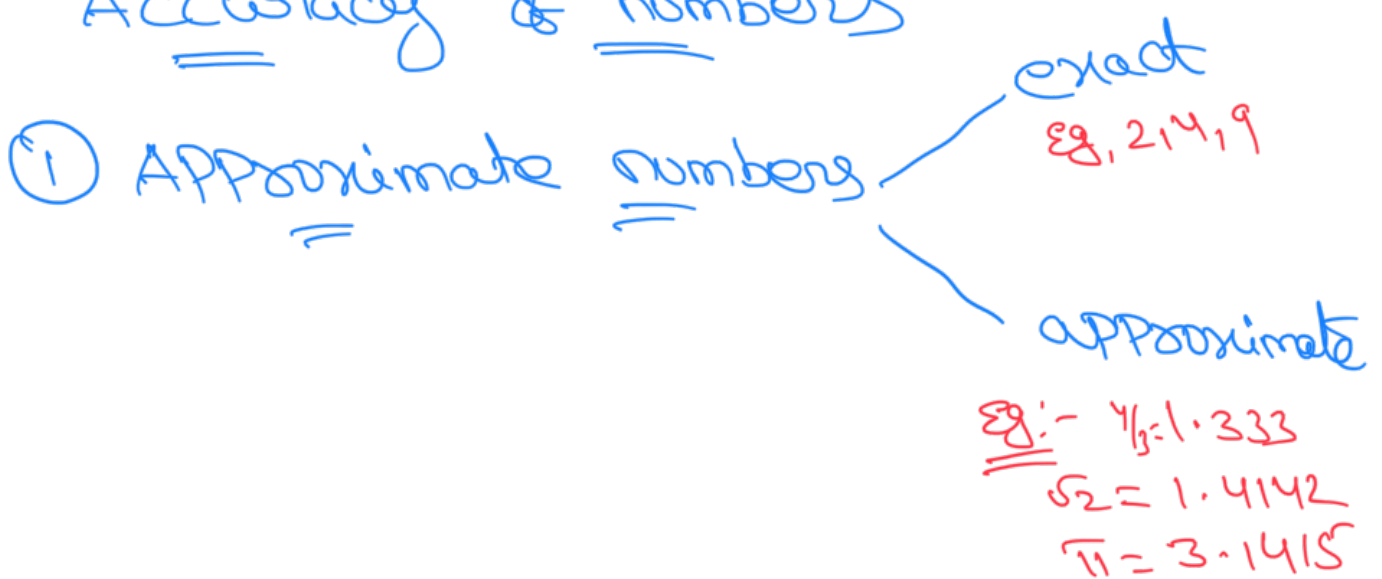
$$\lambda = \frac{1}{2.5a}$$

$\therefore$ The seq generated by this scheme has order of convergence 2 and asymptotic error constant $\frac{1}{2.5a}$

Approximations and Errors in numerical computation

Can refer to class notes
discussed in class
as well from reference
books

Accuracy & numbers



② Significant figures:-

Significant figures

$7845 \rightarrow 4$
 $3.589 \rightarrow 4$
 $0.4758 \rightarrow 4$
 $0.00386 \rightarrow 3$
 $0.000587 \rightarrow 3$
 $45000 \rightarrow 2$
 $7300.00 \rightarrow 2$

③ Round off:-

$$\frac{22}{7} = 3.142857143$$

after round off 3.14 or 3.143

If any number x is given and we want to approximate it by another number x_1 which has n significant digits then two types of approximations are taken

- ① Truncation
- ② Round off

In truncation method n digits of the given number from left is taken and remaining are chopped off.

$X = 23.718928$ truncate to 5 significant digits.

$$\underline{\underline{X_1 = 23.718}}$$

Rule to round off :-

- 1) Discard all digits to the right of the n^{th} digit
- 2) If this discarded number is
 - a) less than half a unit in the n^{th} place, leave the n^{th} digit unchanged
 - b) greater than half a unit in the n^{th} place, increase the n^{th} digit by unity
 - c) exactly half a unit in the n^{th} place, increase the n^{th} digit

by unity if it is odd
otherwise leave it unchanged

→ Following numbers round off upto three significant figures

$$7.893 \rightarrow 7.89$$

$$12.865 \rightarrow 12.9$$

$$6.4356 \rightarrow 6.44$$

$$3.567 \rightarrow 3.57$$

$$84767 \rightarrow 84800$$

$$5.8254 \rightarrow 5.82$$

Absolute, Relative and Percentage Errors

If X is the true value of a quantity and X_1 is its approximate value then $X - X_1$ is called the absolute error

$$E_A = \text{True value} - \text{Approximate Value}$$

$$= X - X_1$$

The relative error is defined

by

$$E_R = \frac{E_A}{X} = \frac{\text{Absolute error}}{\text{True value}}$$

and the Percentage error is

$$E_p = 100 E_R$$

H/W Round off the numbers 865250 and 37.46235 to 4 significant figures and compute E_A , E_R , E_p in each case.

Binary to Decimal

$$\begin{aligned}(11)_2 &= 2^1 \times 1 + 2^0 \times 1 \\ &= 2 + 1 \\ &= 3\end{aligned}$$

$$\begin{aligned}(1101)_2 &= 2^3 \times 1 + 2^2 \times 1 + 0 \times 2^1 + 1 \times 2^0 \\ &= 8 + 4 + 1 \\ &= 13\end{aligned}$$

$$\begin{aligned}(1010)_2 &= 2^3 \times 1 + 2^2 \times 0 + 2^1 \times 1 + 2^0 \times 0 \\ &= 32 + 4 + 1\end{aligned}$$

$$= 37$$

Octal to Binary

$$(20)_8 = (\underline{010} \ \underline{000})_2$$

$$(14)_8 = (\underline{001} \ \underline{100})_2$$

Decimal

0 to 9

and

A B C D E F
↓ ↓ ↓ ↓ ↓ ↓
10 11 12 13 14 15

Binary to Octal

$$(1101)_2 = (\underline{001} \ \underline{101}) = (15)_8$$

Octal to Decimal

$$\begin{aligned} (547)_8 &= 5 \times 8^2 + 4 \times 8^1 + 7 \times 8^0 \\ &= 320 + 32 + 7 \\ &= (359)_{10} \end{aligned}$$

Decimal to Binary

$$(82)_{10} = (1010010)_2$$

$$(109)_{10} = (101101)_2$$

$$\begin{array}{r} 2 \overline{) 82} \\ \underline{41} \\ 2 \overline{) 41} \\ \underline{20} \\ 2 \overline{) 20} \\ \underline{10} \\ 2 \overline{) 10} \\ \underline{5} \\ 2 \overline{) 5} \\ \underline{2} \end{array}$$

$$(3433)_{10} = (1000101001)_2$$

$$\begin{array}{r} 2 \overline{) 2} \rightarrow 1 \\ \underline{1} \rightarrow 0 \end{array}$$

Hexa to Decimal

$$(A3FC)_{16} = 10 \times 16^3 + 3 \times 16^2 + 15 \times 16^1 + 12 \times 16^0$$

$$= 40960 + 768 + 240 + 12 \\ = (41980)_{10}$$

Decimal to Hexa!

$$(2501)_{10} = 091215 \\ = (9C5)_{16}$$

$$\begin{array}{r} 16 \overline{) 2501} \\ 16 \overline{) 186} \rightarrow 15 \\ 16 \overline{) 9} \rightarrow 12 \\ \underline{0} \rightarrow 9 \end{array}$$

for Octa Number

Equivalent Binary num

0	→	000
1	→	001
2	→	010

2	→	0	1	0
3	→	0	1	1
4	→	1	0	0
5	→	1	0	1
6	→	1	1	0
7	→	1	1	1

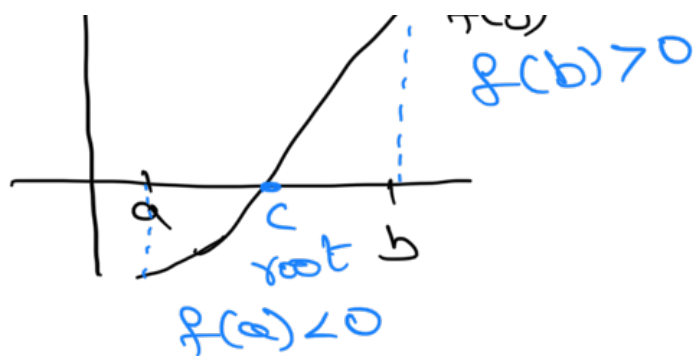
Bisection Method

This method is based on the repeated application of the Intermediate value theorem

Intermediate value theorem:-

If $f(x)$ i.e. a continuous function on some interval $[a, b]$ and $f(a) \cdot f(b) < 0$, then the eq. $f(x) = 0$ has at least one real root or an odd number of real roots in the interval $[a, b]$

1



Algorithm:-

Let the given equation be $f(x)=0$

Step 1:- Find two numbers a and b

such that $f(a) \cdot f(b) < 0$ then the root lies between a and b .

By Bisection method the first approximate value $\frac{a+b}{2} = x_1$

Step 2:- Find $f(x_1)$, $f(x_1) \cdot f(a)$,
 $f(x_1) \cdot f(b)$.

If $f(a) \cdot f(x_1) < 0$ the root lies between a and x_1 . By bisection method the second approximate value $x_2 = \frac{x_1 + a}{2}$

all

$$\text{otherwise } x_2 = \frac{a+TD}{2}$$

Proceed in the above manner
till we get the required
accuracy.

Bisection method: Convergence
Analysis

$f(x)$ is cts on $[a, b]$

$$f(a) \cdot f(b) < 0$$

$$x_1 = \frac{a+b}{2}$$

$$f(x_1) f(a) < 0 \quad (\text{or}) \quad f(x_1) f(b) < 0$$



$$x_1 = \frac{a+b}{2}$$

$$x_1 \in (a, b) = I_0 \Rightarrow |x_1 - p| \leq \frac{b-a}{2}$$

$$x_2 = \frac{a_1+b_1}{2}$$

$$x_2 \in (a_1, b_1) = I_1 \Rightarrow |x_2 - p| \leq \frac{b_1 - a_1}{2}$$

$$x_3 = \frac{a_2+b_2}{2}$$

$$x_3 \in (a_2, b_2) = I_2 \Rightarrow \frac{1}{2} \left(\frac{b-a}{2} \right) = \frac{b-a}{4}$$

$$|x_n - P| \leq \frac{b-a}{2^n}$$

$$|P_n - P| \leq \frac{b-a}{2^n}$$

↓

$$e_n \leq \frac{b-a}{2^n} \rightarrow 0$$

$$|e_n| \rightarrow 0 \text{ as } n \rightarrow \infty$$

Hence, Bisection method
Converges

Order of convergence

We know $|P_n - P| \leq \frac{b-a}{2^n}$

$$|P_n - P| \approx \frac{b-a}{2^n} \quad \text{--- (1)}$$

$$|P_{n+1} - P| \approx \frac{b-a}{2^{n+1}} \quad \text{--- (2)}$$

for n

$$|x_n - P| \leq \frac{b-a}{2^n}$$

$$\leq \frac{1}{2} \left(\frac{b-a}{2^{n-1}} \right)$$

$$\leq \frac{b-a}{2^n}$$

$$|x_3 - P| \leq \frac{b-a}{2^3}$$

$$\frac{b-a}{2^2}$$

o
o
o

o
o
o

$$\frac{(2)}{(1)} \Rightarrow \frac{|P_{n+1}-p|}{|P_n-p|} \approx \frac{(b-a)/2^{n+1}}{(b-a)/2^n} = \frac{1}{2}$$

Here $\alpha = 1$
 $\lambda = 1/2$

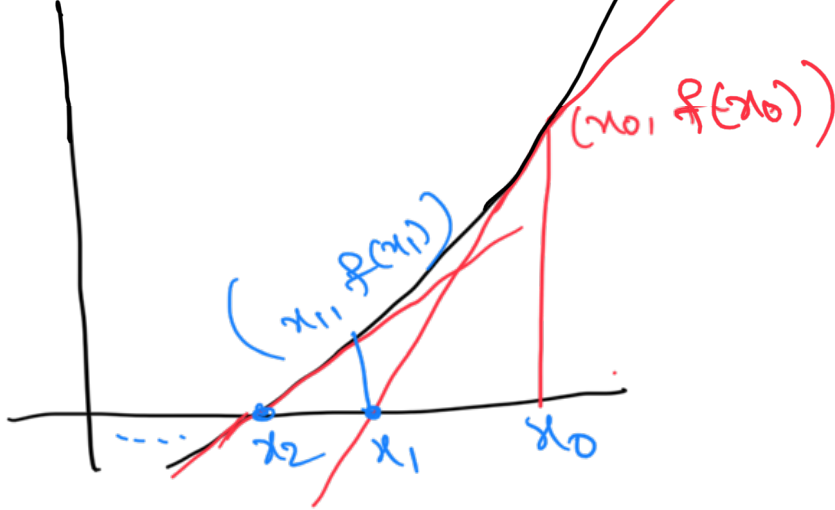
order of convergence of
 Bisection method is 1

NOTE: Asymptotic ^{error} constant $\Rightarrow$ Linear $= 1/2$
 Check class notes for problems

Newton Raphson Method:-

Let x_0 be the initial guess
 for the root of the function
 $f(x)$.

$$y = f(x)$$



Let $P(x_0, f(x_0))$ be a point on the curve corresponding to x_0

By using slope point form

$$\frac{y - f(x_0)}{x - x_0} = f'(x_0)$$

$$y = f(x_0) + f'(x_0)(x - x_0)$$

Now setting $y=0$ and solving for x , we get

$$x = x_0 - \frac{f(x_0)}{f'(x_0)} \quad \text{where } f'(x_0) \neq 0$$

↓
Initial approximation.

Similarly can do further approximations.

In general we have

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

↳ Newton Raphson Formula

Newton Raphson method convergence depends on Initial approximation

Initial approximation is far from root $\Rightarrow$ divergence.

near $\Rightarrow$ convergence.

Secant method:-

we know that

$$f'(x_n) \approx \frac{f(x_n) - f(x_{n-1})}{x_n - x_{n-1}} \rightarrow (1)$$

By Newton Raphson method

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \rightarrow (2)$$

From (1) and (2)

$$\begin{aligned}x_{n+1} &= x_n - \frac{f(x_n)(x_n - x_{n-1})}{f(x_n) - f(x_{n-1})} \\&= \frac{x_n f(x_n) - x_n f(x_{n-1}) - x_n f(x_n) + x_{n-1} f(x_n)}{f(x_n) - f(x_{n-1})} \\&= \frac{x_{n-1} f(x_n) - x_n f(x_{n-1})}{f(x_n) - f(x_{n-1})}\end{aligned}$$

(3) is the formula (3)

for finding root using

Second method

i.e

$$x_{n+1} = \frac{x_{n-1} f(x_n) - x_n f(x_{n-1})}{f(x_n) - f(x_{n-1})}$$

Second
Formula ✓✓

$$x_2 = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)}$$

two initial
guesses needed

i.e. $x_0 = \underline{\hspace{2cm}}$

$x_1 = \underline{\hspace{2cm}}$

NOTE!

Bisection	Newton Raphson	Secant
$f(x) = 0$	$f(x) = 0$	$f(x) = 0$
no initial guess	Initial guess $x_0 = 0$	Two initial $x_0 = \underline{\hspace{2cm}}$ $x_1 = \underline{\hspace{2cm}}$
$f(a) \cdot f(b) < 0$		

Accuracy up to 4 decimal
places

NOTE:-

DO discussed problems
on Bisection, Secant and
Newton Raphson Method
in the class

- 1) Find a root of $x^3 - 5x + 3 = 0$
using Newton Raphson method
- 2) Find the root of $x e^{-2x} = \frac{\sin x}{2}$
- 3) Find the value of $\sqrt{7}$
- 4) Find a root of $x^3 + 3x^2 - 3 = 0$
- 5) Find a real root of equation
 $x^3 - 2x - 5 = 0$. Using Secant
method. Taking the initial
approximations as 2 and 3
- 6) Find a root of $x^3 - x - 1 = 0$
- 7) Find a root of $x^3 - x - 4 = 0$
- 8) Find a root of $x^3 + x^2 + x + 7 = 0$